

A ()

A ()

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A ()

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A

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- Python
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A ():

A

(r, θ)
 $\vec{r}$

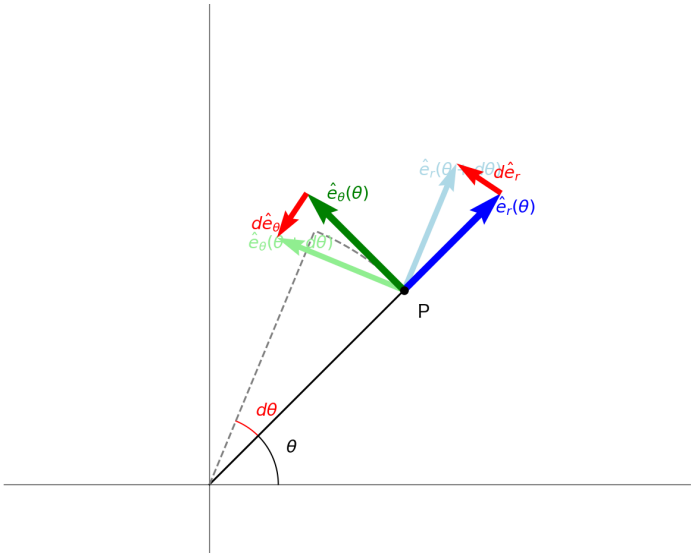
$\hat{e}_r$
 $\hat{e}_\theta$

$(\hat{i}, \hat{j})$
 $\hat{e}_r, \hat{e}_\theta$

$\theta \quad \theta + d\theta$
 $d\hat{e}_r, d\hat{e}_\theta$

$\vec{r} = r\hat{e}_r$

極座標の単位ベクトルの微分（定性的な図解）



1:

$$\begin{array}{ccccccc}
& & 1. \frac{d\hat{e}_r}{d\hat{e}_\theta} : \hat{e}_r & \frac{d\theta}{\hat{e}_r} & d\hat{e}_r & \hat{e}_\theta & 1 \cdot d\theta \\
d\theta & & & & 1 \cdot d\theta & & 2. \frac{d\hat{e}_\theta}{d\hat{e}_r} : \hat{e}_\theta \\
& dt & & & & & \\
& & & \frac{d\hat{e}_r}{dt} = \dot{\theta}\hat{e}_\theta, & \frac{d\hat{e}_\theta}{dt} = -\dot{\theta}\hat{e}_r & &
\end{array}$$


```
#| standalone: true
#| components: [editor, viewer]

import numpy as np
import matplotlib.pyplot as plt
from shiny import App, render, ui

# --- UI ---
app_ui = ui.page_fluid(
  ui.layout_sidebar(
    ui.sidebar(
      ui.input_slider("length", "    L (m)", 0.1, 2.0, 1.0, step=0.1),
      ui.input_slider("initial_angle", "    0 (rad)", -np.pi/2, np.pi/2, np.pi/4, step=
      ui.input_slider("gravity", "    g (m/s²)", 1.0, 20.0, 9.8, step=0.1),
      #
      ui.input_slider("time", "    t (s)", 0.0, 10.0, 0.0, step=0.1,
                      animate=ui.AnimationOptions(interval=100, loop=True))
    ),
    ui.output_plot("pendulum_plot"),
  )
)

# --- ---
def server(input, output, session):
  @render.plot(alt="    ")
  def pendulum_plot():
```

```

#
L = input.length()
g = input.gravity()
theta0 = input.initial_angle()
t = input.time()

#
omega = np.sqrt(g / L)
theta = theta0 * np.cos(omega * t)

# (x, y)
x = L * np.sin(theta)
y = -L * np.cos(theta)

#
fig, ax = plt.subplots(figsize=(6, 6))
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(-2.5, 0.5)
ax.set_aspect('equal')
ax.grid(True)
ax.set_title(f"  t = {t:.1f} s          ")

#
ax.plot(0, 0, 'ko', markersize=8)          #
ax.plot([0, x], [0, y], 'k-', lw=2)        #
ax.plot(x, y, 'ro', markersize=20)         #

return fig

#
app = App(app_ui, server)

```

l m

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l} \sin \theta$$

$\theta \ll 1$ $\sin \theta \approx \theta$

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta$$

$$\theta(t) = \theta_0 \cos\left(\sqrt{\frac{g}{l}}t + \phi\right)$$

$$r = ae^{k\theta} \ (\theta > 0) \quad \dot{\theta} = \omega = \text{const} \quad v_r, v_\theta, a_r, a_\theta$$

$$a > 0 \quad a_r \quad k$$

k

```
#| standalone: true
#| components: [editor, viewer]

import numpy as np
import matplotlib.pyplot as plt
from shiny import App, render, ui

# --- UI ---
app_ui = ui.page_fluid(
    ui.layout_sidebar(
        ui.sidebar(
            ui.input_slider("k", "    k", -0.5, 0.5, 0.15, step=0.05),
            ui.input_slider("omega", "    ", 0.5, 5.0, 1.5, step=0.1),
            ui.input_slider("t", "    t (s)", 0.0, 5.0, 0.0, step=0.1,
                           animate=ui.AnimationOptions(interval=100, loop=True))
        ),
        ui.output_plot("spiral_plot"),
    )
)

# ---
```

```

def server(input, output, session):
    @render.plot(alt=" ")
    def spiral_plot():
        k = input.k()
        omega = input.omega()
        t = input.t()
        a = 1.0 #

        #
        #
        theta_max = omega * t
        if theta_max == 0:
            theta_max = 0.01 #

        theta_array = np.linspace(0, theta_max, 200)

        # (r, theta) (x, y)
        r_array = a * np.exp(k * theta_array)
        x_array = r_array * np.cos(theta_array)
        y_array = r_array * np.sin(theta_array)

        #
        x_current = x_array[-1]
        y_current = y_array[-1]

        # --- ---
        fig, ax = plt.subplots(figsize=(6, 6))

        #
        max_r = max(a, a * np.exp(k * omega * 5.0))
        ax.set_xlim(-max_r*1.1, max_r*1.1)
        ax.set_ylim(-max_r*1.1, max_r*1.1)
        ax.set_aspect('equal')
        ax.grid(True, linestyle='--', alpha=0.7)
        ax.set_title(f" ( t = {t:.1f})")

        #
        ax.plot(x_array, y_array, 'k-', alpha=0.5, label=' ')
        #
        ax.plot([0, x_current], [0, y_current], 'b-', lw=1.5, label='$\\vec{r}$')
        #
        ax.plot(x_current, y_current, 'ro', markersize=8)

```

```

ax.legend(loc='upper left')
return fig

#
app = App(app_ui, server)

```

1.

$$\vec{r} = r\hat{e}_r \quad t$$

$$(AB)' = A'B + AB'$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{e}_r) = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$v_r = \dot{r}, v_\theta = r\dot{\theta}$$

$$3 \quad (ABC)' = A'BC + AB'C + ABC'$$

$$\begin{aligned} \vec{a} &= \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{e}_r) + \frac{d}{dt}(r\dot{\theta}\hat{e}_\theta) \\ &= (\ddot{r}\hat{e}_r + \dot{r}\dot{\hat{e}}_r) + (\dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\dot{\hat{e}}_\theta) \end{aligned}$$

$$\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta \quad \dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r$$

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

$$a_r, a_\theta$$

2.

$$\begin{aligned}
 * r &= ae^{k\theta} * \dot{\theta} = \omega \quad () * \ddot{\theta} = \frac{d}{dt}(\omega) = 0 \\
 r & \quad \dot{r} \quad \theta \quad t \quad r(t) = ae^{k\theta(t)} \quad t \quad \theta \quad \dot{\theta} \\
 \dot{r} &= \frac{dr}{d\theta} \cdot \frac{d\theta}{dt} = \frac{d}{d\theta}(ae^{k\theta}) \cdot \omega = (ake^{k\theta}) \cdot \omega = ak\omega e^{k\theta} \\
 ae^{k\theta} &= r \\
 \dot{r} &= k\omega r
 \end{aligned}$$

$$\begin{aligned}
 \ddot{r} &= \frac{d}{dt}(C \cdot f(t)) = C \cdot \dot{f}(t) \\
 \ddot{r} &= \frac{d}{dt}(k\omega r) = k\omega \cdot \dot{r} \\
 \dot{r} &= ak\omega e^{k\theta} \\
 \ddot{r} &= k\omega(ak\omega e^{k\theta}) = ak^2\omega^2 e^{k\theta}
 \end{aligned}$$

:

$$\begin{aligned}
 v_r &= \dot{r} = ak\omega e^{k\theta} \\
 v_\theta &= r\dot{\theta} = (ae^{k\theta})\omega = a\omega e^{k\theta}
 \end{aligned}$$

:

$$\begin{aligned}
 a_r &= \ddot{r} - r\dot{\theta}^2 = ak^2\omega^2 e^{k\theta} - (ae^{k\theta})\omega^2 = a\omega^2(k^2 - 1)e^{k\theta} \\
 a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} = (ae^{k\theta})(0) + 2(ak\omega e^{k\theta})\omega = 2ak\omega^2 e^{k\theta}
 \end{aligned}$$

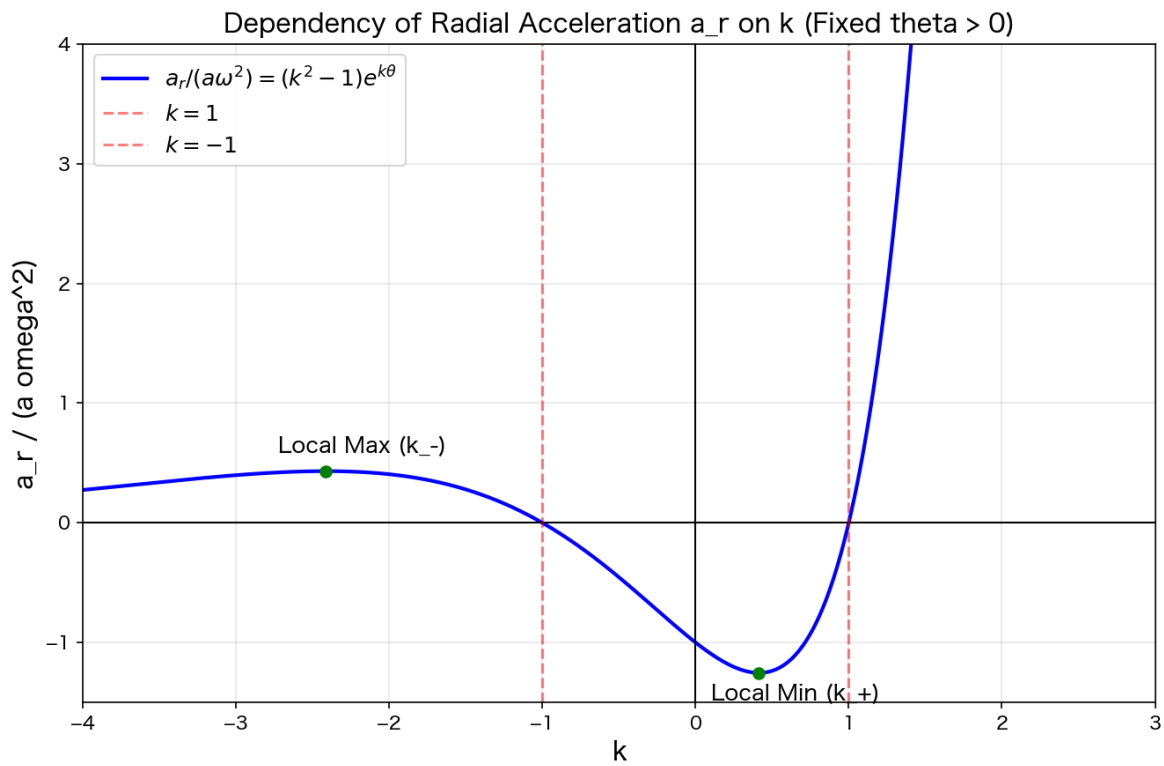
3. $a_r = k$

$$\begin{aligned}
 a_r(k) &= a\omega^2(k^2 - 1)e^{k\theta} \quad a > 0 \quad \theta > 0 \quad k \quad a\omega^2 > 0 \\
 f(k) &= (k^2 - 1)e^{k\theta}
 \end{aligned}$$

$$\begin{aligned}
 f'(k) &= 2ke^{k\theta} + (k^2 - 1) \cdot \theta e^{k\theta} = (\theta k^2 + 2k - \theta)e^{k\theta} \\
 f'(k) &= 0 \quad k = 2 \quad \theta k^2 + 2k - \theta = 0
 \end{aligned}$$

$$k_{\pm} = \frac{-1 \pm \sqrt{1 + \theta^2}}{\theta}$$

$$\theta > 0 \quad k_+ > 0 \quad k_- < -1$$



1: a_r k

1. $a_r : e^{k\theta} \quad a_r (k^2 - 1)$

- $|k| < 1 : a_r < 0$
- $|k| > 1 : a_r > 0$
- $k = \pm 1 : a_r = 0$

2. $k \rightarrow \infty : e^{k\theta}$

3. $k \rightarrow -\infty : (k^2 - 1) e^{k\theta} \rightarrow 0 \quad a_r \rightarrow 0$

4. $k < -1 : 0 < k < 1$

0